

Notes: Helpman, Melitz, and Rubinstein (QJE, 2008)

Estimating Trade Flows: Trading Partners and Trading Volumes

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1 Big picture

- **Question.** How should researchers estimate gravity equations when many country pairs do not trade and when the number of exporters varies across destinations?
- **Core problem.** Standard gravity regressions usually use only positive bilateral trade flows. This drops the information in zero trade flows and also ignores the extensive margin - the number or fraction of firms that select into exporting to a destination.
- **Theoretical setting.** The paper builds a heterogeneous-firm trade model with fixed and variable export costs. Only sufficiently productive firms export to a destination, and some country pairs have no exporters at all.
- **Main empirical object.** The paper estimates bilateral trade flows using 158 countries and directed country-pair observations. The model jointly explains whether a country pair trades and, conditional on trading, how much it trades.
- **Key decomposition.** The effect of trade frictions is decomposed into an **intensive margin**, the trade volume per exporter, and an **extensive margin**, the number or fraction of exporting firms.
- **Main empirical contribution.** The authors develop a two-stage estimator: a first-stage selection equation for whether country j exports to country i , followed by a second-stage gravity equation for positive trade flows that corrects for selection and unobserved firm heterogeneity.
- **Main result.** Traditional gravity estimates overstate the direct firm-level effect of trade frictions. Most of the bias comes from omitting the extensive margin, not from ordinary sample selection.
- **Economic implication.** Trade barriers do not merely reduce shipments by existing exporters. They also reduce the set of firms that export. Ignoring this extensive-margin channel confounds product-level trade responses with exporter entry responses.
- **Broader takeaway.** Aggregate trade data can reveal information about firm selection into export markets, even without firm-level data, if the selection equation and trade-flow equation are estimated jointly.

2 Incremental contribution of the paper

- **From gravity with positive flows to gravity with zeros.** Earlier gravity work often discarded zero trade flows. This paper shows that zeros are theoretically meaningful outcomes generated by fixed export costs and heterogeneous firms.
- **A bridge between Melitz-style firm heterogeneity and aggregate gravity estimation.** The paper brings the heterogeneous-firm logic of exporter selection into a gravity equation that can be estimated with country-level data.
- **A two-equation empirical system.** The paper does not simply add a correction term mechanically. It derives both the selection equation and the trade-flow equation from the same model.
- **A decomposition of trade-friction effects.** The paper separates the effect of trade frictions on trade per exporter from their effect on the fraction of exporters.

- **A bias diagnosis.** The paper shows that standard gravity coefficients are biased mainly because the extensive margin is omitted. The Heckman-style selection correction alone leaves most benchmark estimates almost unchanged.
- **A flexible estimation contribution.** The results are shown to be robust across parametric nonlinear least squares, semiparametric polynomial approximation, and nonparametric bin approaches.
- **A new use of aggregate data.** The model allows the researcher to infer sufficient statistics for firm-level exporter selection from aggregate bilateral flows and country-pair characteristics.
- **A heterogeneity result.** The bias in standard gravity estimation is not uniform. Trade-cost elasticities differ sharply across country pairs, especially across North-North, North-South, and South-South pairs.

3 Data and measurement

3.1 Bilateral trade data

- The bilateral trade data come from Feenstra’s world trade flow data sets covering 1970 to 1997.
- The usable sample contains 158 countries.
- Directed trade-flow observations are constructed as exports from country j to country i .
- With 158 countries, there are $158 \times 157 = 24,806$ possible directed bilateral trade relationships in a given year.
- The main cross-sectional analysis uses 1986, and the paper also checks patterns over the 1980s.
- Trade flows are expressed in constant 2000 U.S. dollars.

Interpretation: The unit of observation is directional: exports from j to i . This matters because the model allows M_{ij} and M_{ji} to differ and even allows trade in only one direction.

3.2 Zero trade and country-pair status

- Country pairs can trade in both directions, in one direction only, or not at all.
- Roughly half of country-pair observations involve no trade or one-way trade.
- The data therefore contain much more information than the sample of positive two-way trade flows used in many traditional gravity exercises.
- The paper uses this information to estimate the probability that a country pair trades.

Interpretation: Zero trade flows are not treated as missing observations. They are equilibrium outcomes that reveal information about fixed trade costs and exporter selection.

3.3 Country-pair gravity variables

The paper uses standard gravity variables that proxy for variable and fixed trade frictions:

- log geographic distance between capitals,
- common land border,
- both countries being islands,
- both countries being landlocked,
- common legal origin,
- common language,
- colonial ties,
- common currency union,
- common regional free trade agreement,
- common religion index,
- GATT/WTO membership variables in the 1980s specifications.

Interpretation: Many of these variables affect both whether countries trade and how much they trade. This is the empirical reason standard positive-flow gravity regressions can be biased.

3.4 Excluded variables for the selection equation

- The first excluded-variable strategy uses country-level regulation costs of firm entry from Djankov et al. (2002).
- These variables measure the number of days, procedures, and relative costs required to start a business.
- The authors construct indicators for country pairs in which both importer and exporter have high entry regulation costs.
- The argument is that these variables affect fixed costs of exporting, and hence the probability of trade, but not variable costs per unit of exports.
- Because entry regulation data are unavailable for many countries, this reduced-sample approach loses many observations.
- To use the full sample, the authors test and then use common religion as an alternative excluded variable.

Interpretation: The exclusion restriction is central: the excluded variable should shift the fixed cost of becoming a trading partner but not the volume shipped by firms conditional on exporting.

4 Models and empirical specifications

4.1 Heterogeneous-firm trade model

Countries consume differentiated products with CES preferences. Country j has N_j firms, each producing a distinct variety. Firms differ in productivity. A firm with unit input coefficient a has productivity $1/a$, where lower a means higher productivity.

A country- j firm selling to country i faces:

- a variable iceberg trade cost τ_{ij} ,
- a fixed export cost $c_j f_{ij}$,
- monopolistic-competition pricing with constant markup.

Only firms with productivity high enough to cover the fixed cost export to a destination. The cutoff productivity condition is summarized by a threshold a_{ij} satisfying the zero-profit condition.

Interpretation: The model naturally generates three observed outcomes: two-way trade, one-way trade, and zero trade. A country can export to some destinations but not others because fixed costs and demand conditions differ by destination.

4.2 Export cutoff and extensive margin

Let $G(a)$ be the distribution of firm productivity draws. Only firms with $a \leq a_{ij}$ export from j to i . The fraction of firms exporting from j to i is therefore $G(a_{ij})$.

The aggregate trade volume from j to i can be written as

$$M_{ij} = \left(\frac{c_j \tau_{ij}}{\alpha P_i} \right)^{1-\epsilon} Y_i N_j V_{ij}, \quad (1)$$

where V_{ij} aggregates the productivity of the firms from j that export to i .

- If no firm can profitably export, then $V_{ij} = 0$ and $M_{ij} = 0$.
- If some firms export, then V_{ij} rises with the fraction of exporters.
- Thus the same trade friction affects both firm-level export volumes and the number of firms exporting.

Interpretation: The omitted variable in a standard gravity equation is essentially V_{ij} , or a transformation of the fraction of exporting firms. This is the extensive-margin correction.

4.3 Log-linear trade-flow equation

The model implies the positive-flow equation

$$m_{ij} = \beta_0 + \lambda_j + \chi_i - \gamma d_{ij} + w_{ij} + u_{ij}, \quad (2)$$

where:

- m_{ij} is log exports from country j to country i ,
- λ_j is an exporter fixed effect,
- χ_i is an importer fixed effect,
- d_{ij} is log distance or a vector of trade barriers,
- w_{ij} captures the extensive margin, that is, the fraction of exporting firms,

- u_{ij} is an unobserved bilateral trade-cost shock.

Interpretation: If w_{ij} is omitted, the distance coefficient does not measure only the effect of distance on trade per exporter. It also picks up the effect of distance on the number of exporters.

4.4 Selection equation

The probability of positive exports is derived from the profitability of the most productive firm. Define Z_{ij} as the ratio of variable export profits for the most productive firm to the fixed export cost of exporting. Positive exports occur when $Z_{ij} > 1$.

The latent selection equation is

$$z_{ij} = \gamma_0 + \xi_j + \zeta_i - \gamma d_{ij} - \kappa \phi_{ij} + \eta_{ij}, \quad (3)$$

where ϕ_{ij} represents observed fixed-trade-cost variables. The estimable first-stage Probit is

$$\rho_{ij} = Pr(T_{ij} = 1 \mid \text{observables}) = \Phi \left(\gamma_0^* + \xi_j^* + \zeta_i^* - \gamma^* d_{ij} - \kappa^* \phi_{ij} \right). \quad (4)$$

- $T_{ij} = 1$ if country j exports to country i .
- ρ_{ij} is the predicted probability of positive exports.
- The predicted latent index from this first stage is used to construct both the selection correction and the extensive-margin correction.

Interpretation: The first stage is not an auxiliary prediction exercise. In the model, it identifies the cutoff for the marginal exporter and therefore the extensive margin in the trade-flow equation.

4.5 Second-stage correction

The second-stage equation includes two generated controls:

- an inverse-Mills-ratio style term, $\hat{\eta}_{ij}^*$, which corrects for sample selection into positive trade flows;
- an extensive-margin term, constructed from the predicted latent profitability index, which corrects for the number of exporters.

In the parametric version, the paper estimates a nonlinear second-stage equation of the form

$$m_{ij} = \beta_0 + \lambda_j + \chi_i - \gamma d_{ij} + \ln \left\{ \exp \left[\delta (\hat{z}_{ij}^* + \hat{\eta}_{ij}^*) \right] - 1 \right\} + \beta_{un} \hat{\eta}_{ij}^* + e_{ij}. \quad (5)$$

Interpretation: The paper's key empirical innovation is that a standard Heckman correction is not enough. One must also correct for the endogenous extensive margin generated by heterogeneous firms.

4.6 Parametric, semiparametric, and nonparametric versions

- **Parametric NLS:** assumes productivity follows a truncated Pareto distribution and uses the implied nonlinear extensive-margin function.

- **Polynomial approximation:** relaxes the Pareto assumption and approximates the extensive-margin function with a polynomial in the predicted latent index.
- **Indicator-variable bins:** relaxes more distributional assumptions by using many bins of the predicted trade probability.

Interpretation: The important empirical result is that all three approaches produce similar corrections. The main findings are therefore not driven by one strict functional-form assumption.

5 Identification and empirical design

5.1 What the design can identify

- The first stage identifies how country-pair characteristics affect the probability that one country exports to another.
- The second stage identifies how trade barriers affect positive trade flows after correcting for selection and the exporter extensive margin.
- The model allows aggregate data to infer the extensive margin even though the data do not directly observe every exporting firm.

5.2 Why exporter and importer fixed effects matter

- The paper uses separate exporter and importer fixed effects, rather than a single country-pair symmetric setup.
- Exporter fixed effects absorb exporter productivity, cost, and number-of-firms components.
- Importer fixed effects absorb importer expenditure, price-index, and market-size components.
- This directional specification is necessary because trade flows can be asymmetric.

Interpretation: The model gives a theoretical justification for using directed trade flows and separate exporter and importer fixed effects.

5.3 Exclusion restriction logic

- The first-stage excluded variables should affect the fixed cost of forming a trade relationship.
- They should not directly affect the variable cost or trade volume of firms that already export.
- Entry regulation costs fit this logic because they raise fixed operating costs.
- Common religion is used after testing that it predicts selection but is not correlated with second-stage residuals in the reduced sample.

Interpretation: The exclusion restriction is plausible, not automatic. The paper supports it theoretically for regulation costs and empirically for religion.

5.4 Main threats and how the paper addresses them

- **Sample selection.** Dropping zero trade flows may bias gravity estimates. The paper uses a Probit selection equation and inverse-Mills-ratio term.
- **Omitted extensive margin.** Positive trade flows depend on how many firms export. The paper adds an extensive-margin control derived from the first stage.
- **Functional-form dependence.** The paper checks NLS, polynomial, and bin-based specifications.
- **Asymmetry.** Bilateral flows can differ by direction. The paper uses directional trade flows and separate fixed effects.
- **External validation.** The estimated extensive margin for French exports is highly correlated with the number of French exporters by destination reported in firm-level data.

Interpretation: The paper’s credibility comes from combining theory, selection-equation evidence, alternative excluded variables, flexible estimation, and external validation using firm-level exporter counts.

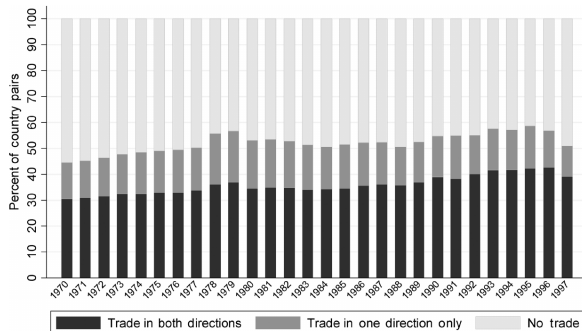
6 Tables and figures

6.1 Figures I and II: Zero trade and growth of aggregate trade

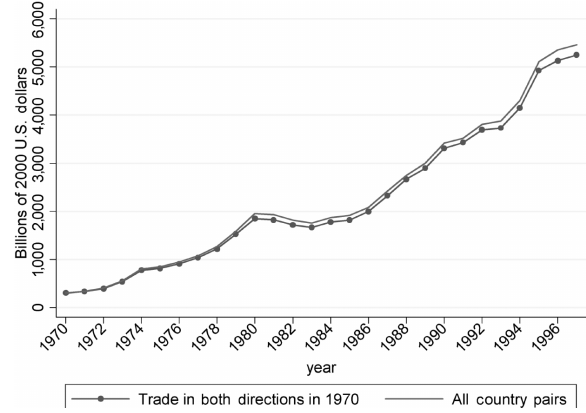
Read:

- Figure I classifies country pairs by whether they trade in both directions, one direction only, or not at all.
- A large share of country pairs do not trade or trade only one way, so positive-flow gravity regressions discard substantial information.
- Figure II shows that the growth of aggregate exports from 1970 to 1997 came mostly from country pairs that already traded in both directions in 1970.
- New country-pair trade relationships contributed relatively little to aggregate world trade growth over this period.

Economic message: The figures motivate the full empirical approach. The many zeros make selection important, while the growth pattern shows that the extensive margin within existing trading relationships - more firms or products per partner - matters for trade volumes.



(a) Figure I: Country-pair trade status



(b) Figure II: Aggregate exports and initial trading pairs

6.2 Tables I and II: Benchmark gravity, selection, and baseline corrections

Read:

- Table I reports traditional gravity estimates and first-stage Probit estimates for trade selection.
- Standard gravity variables such as distance, language, legal origin, currency union, FTA, and colonial ties affect positive trade volumes.
- The Probit columns show that many of the same variables also affect whether a trade relationship exists.
- This is direct evidence that selection into trading partners is not random.
- Table II implements the two-stage estimator on the reduced sample using entry-regulation costs as excluded variables.
- The distance coefficient falls in magnitude from about -1.17 in the benchmark to roughly -0.8 in the corrected specifications.
- Language, religion, and FTA effects become much smaller or insignificant in corrected specifications, suggesting that they mainly affect fixed costs and exporter entry.

Economic message: Table I shows why a correction is needed; Table II shows what changes after correcting. The benchmark gravity equation mixes intensive-margin effects with extensive-margin effects.

TABLE I
BENCHMARK GRAVITY AND SELECTION INTO TRADING RELATIONSHIPS

Variables	1986			1980s		
	(Probit)		m_{ij}	(Probit)		T_{ij}
	m_{ij}	T_{ij}		m_{ij}	T_{ij}	
Distance	-1.176** (0.031)	-0.263** (0.012)	-1.201** (0.024)	-0.246** (0.008)	-1.200** (0.024)	-0.246** (0.008)
Land border	0.458** (0.147)	-0.148** (0.047)	0.366** (0.131)	-0.146** (0.032)	0.364** (0.131)	-0.146** (0.032)
Island	-0.391** (0.121)	-0.136** (0.032)	-0.381** (0.096)	-0.140** (0.022)	-0.378** (0.096)	-0.140** (0.022)
Landlock	-0.561** (0.188)	-0.072 (0.045)	-0.582** (0.148)	-0.087** (0.028)	-0.581** (0.147)	-0.087** (0.028)
Legal	0.486** (0.050)	0.038** (0.014)	0.406** (0.040)	0.029** (0.009)	0.407** (0.040)	0.028** (0.009)
Language	0.176** (0.061)	0.113** (0.016)	0.207** (0.047)	0.109** (0.011)	0.203** (0.047)	0.108** (0.011)
Colonial ties	1.299** (0.120)	0.128 (0.117)	1.321** (0.110)	0.114 (0.082)	1.326** (0.110)	0.116 (0.082)
Currency union	1.364** (0.255)	0.190** (0.052)	1.395** (0.187)	0.206** (0.026)	1.409** (0.187)	0.206** (0.026)
FTA	0.759** (0.222)	0.494** (0.020)	0.996** (0.213)	0.497** (0.018)	0.976** (0.214)	0.495** (0.018)
Religion	0.102 (0.096)	0.104** (0.025)	-0.018 (0.076)	0.099** (0.016)	-0.038 (0.077)	0.098** (0.016)
WTO (none)					-0.068 (0.058)	-0.056** (0.013)
WTO (both)					0.303** (0.042)	0.093** (0.013)
Observations	11,146	24,649	110,697	248,060	110,697	248,060
R^2	0.709	0.587	0.682	0.551	0.682	0.551

Notes. Exporter, importer, and year fixed effects. Marginal effects at sample means and pseudo R^2 reported for Probit. Robust standard errors (clustering by country pair).

(a) Table I: Benchmark gravity and selection

TABLE II
BASELINE RESULTS

Variables	1986 reduced sample					
	m_{ij}					Indicator variables
	(Probit) T_{ij}	Benchmark	NLS	Polynomial	50 bins	100 bins
Distance	-0.213** (0.016)	-1.167** (0.040)	-0.813** (0.049)	-0.847** (0.052)	-0.755** (0.070)	-0.789** (0.088)
Land border	-0.087 (0.072)	0.627** (0.165)	0.871** (0.170)	0.845** (0.166)	0.892** (0.170)	0.863** (0.170)
Island	-0.173* (0.078)	-0.553* (0.269)	-0.203 (0.290)	-0.218 (0.258)	-0.161 (0.259)	-0.197 (0.258)
Landlock	-0.053 (0.050)	-0.432* (0.189)	-0.347* (0.175)	-0.362+ (0.187)	-0.352+ (0.187)	-0.353+ (0.187)
Legal	0.049** (0.019)	0.535** (0.064)	0.431** (0.065)	0.434** (0.064)	0.407** (0.065)	0.418** (0.065)
Language	0.101** (0.021)	0.147+ (0.075)	-0.030 (0.087)	-0.017 (0.077)	-0.061 (0.079)	-0.036 (0.083)
Colonial ties	-0.009 (0.130)	0.909** (0.158)	0.847** (0.257)	0.848** (0.148)	0.853** (0.152)	0.838** (0.153)
Currency union	0.216** (0.038)	1.534** (0.334)	1.077** (0.360)	1.150** (0.333)	1.045** (0.337)	1.107** (0.346)
FTA	0.343** (0.009)	0.976** (0.247)	0.124 (0.227)	0.241 (0.197)	-0.141 (0.250)	0.065 (0.348)
Religion	0.141** (0.034)	0.281* (0.120)	0.120 (0.136)	0.139 (0.120)	0.073 (0.124)	0.100 (0.128)
Regulation costs	-0.108** (0.036)	-0.146 (0.100)				
R. costs (days & proc.)	-0.061* (0.031)	-0.216+ (0.124)				
δ (from $\hat{w}_{ij}^*$)			0.840** (0.043)			
$\hat{\eta}_{ij}^*$			0.240* (0.099)	0.882** (0.209)		
$\hat{z}_{ij}^*$				3.261** (0.540)		
$\hat{z}_{ij}^{*2}$				-0.712** (0.170)		
$\hat{z}_{ij}^{*3}$				0.060** (0.017)		
Observations	12,198	6,602	6,602	6,602	6,602	6,602
R^2	0.573	0.693		0.701	0.704	0.706

Notes. Exporter and importer fixed effects. Marginal effects at sample means and pseudo R^2 reported for Probit. Regulation costs are excluded variables in all second stage specifications. Bootstrapped standard errors for NLS; robust standard errors (clustering by country pair) elsewhere.
+Significant at 10%.

(b) Table II: Baseline two-stage results

6.3 Table III: Alternative excluded variables and full-sample estimates

Read:

- Table III replaces entry-regulation excluded variables with common religion and reports estimates for both the reduced sample and the full sample.
- The estimates are very similar to Table II, supporting the use of religion as an alternative excluded variable.
- The full-sample results roughly double the number of positive trade observations, from 6,602 to 11,146.
- Corrected distance coefficients remain much smaller in magnitude than the benchmark gravity coefficient.
- FTA, language, and related variables again show large declines relative to benchmark estimates.

Economic message: The main conclusions are not driven by the smaller regulation-cost sample. Using a larger sample and a different excluded variable produces the same qualitative message: the extensive margin matters and standard gravity estimates overstate direct trade-friction effects.

TABLE III
ALTERNATE EXCLUDED VARIABLES

Variables	1986 reduced sample				1986 full sample			
	NLS	Polynomial	Indicator variables		NLS	Polynomial	Indicator variables	
			50 bins	100 bins			50 bins	100 bins
Distance	-0.822** (0.048)	-0.853** (0.051)	-0.751** (0.069)	-0.731** (0.089)	-0.798** (0.039)	-0.862** (0.041)	-0.671** (0.059)	-0.623** (0.076)
Land border	0.878** (0.169)	0.855** (0.164)	0.903** (0.166)	0.907** (0.167)	0.834** (0.132)	0.786** (0.144)	0.894** (0.147)	0.924** (0.150)
Island	-0.204 (0.291)	-0.219 (0.258)	-0.171 (0.265)	-0.142 (0.266)	-0.169 (0.120)	-0.200* (0.118)	-0.091 (0.119)	-0.074 (0.121)
Landlock	-0.348* (0.176)	-0.361+ (0.188)	-0.347+ (0.190)	-0.344+ (0.192)	-0.447** (0.172)	-0.482** (0.186)	-0.437* (0.186)	-0.439* (0.186)
Legal	0.439** (0.066)	0.442** (0.064)	0.424** (0.065)	0.418** (0.066)	0.387** (0.048)	0.385** (0.049)	0.350** (0.050)	0.345** (0.050)
Language	-0.018 (0.085)	-0.005 (0.077)	-0.060 (0.079)	-0.068 (0.085)	0.023 (0.062)	0.045 (0.061)	-0.044 (0.064)	-0.062 (0.068)
Colonial ties	0.835** (0.251)	0.839** (0.147)	0.837** (0.149)	0.830** (0.148)	1.001** (0.204)	1.038** (0.116)	0.960** (0.117)	0.929** (0.119)
Currency union	1.034** (0.361)	1.102** (0.334)	1.021** (0.341)	0.984** (0.353)	1.023** (0.273)	1.106** (0.261)	0.977** (0.265)	0.960** (0.270)

(a) Table III, part 1: Alternative excluded variables

TABLE III
(CONTINUED)

Variables	1986 reduced sample				1986 full sample			
	NLS	Polynomial	Indicator variables		NLS	Polynomial	Indicator variables	
			50 bins	100 bins			50 bins	100 bins
FTA	0.154 (0.225)	0.254 (0.199)	-0.161 (0.250)	-0.200 (0.337)	0.380* (0.182)	0.457** (0.162)	0.050 (0.165)	-0.091 (0.210)
$\hat{\beta}_{ij}$ (from $\hat{\beta}_{ij}^*$)	0.827** (0.043)				0.871** (0.028)			
$\hat{\beta}_{ij}^*$	0.198* (0.099)	0.823** (0.211)			0.372** (0.069)	1.131** (0.138)		
$\hat{\beta}_{ij}^*$		3.229** (0.538)				3.802** (0.386)		
$\hat{\beta}_{ij}^*$		-0.709** (0.189)				-0.782** (0.123)		
$\hat{\beta}_{ij}^*$		0.061** (0.017)				0.064** (0.013)		
Observations	6,602	6,602	6,602	6,602	11,146	11,146	11,146	11,146
R^2	0.700	0.702	0.705		0.721	0.722	0.723	

Notes. m_{ij} is dependent variable throughout. Exporter and importer fixed effects. Religion is excluded variable in all second stage specifications. Bootstrapped standard errors for NLS; robust standard errors (clustering by country pair) elsewhere.
*Significant at 10%.
**Significant at 5%.
***Significant at 1%.

(b) Table III, part 2: Alternative excluded variables, continued

6.4 Tables IV and V: Bias decomposition and asymmetries

Read:

- Table IV decomposes the bias in traditional gravity estimates into a firm-heterogeneity correction and a Heckman selection correction.
- Adding only the firm-heterogeneity or extensive-margin control reproduces most of the corrected results.
- Adding only the Heckman selection correction leaves the benchmark coefficients largely unchanged.
- This shows that the economically important bias comes from omitting the extensive margin, not from selection alone.
- Table V studies trade-flow asymmetries.
- Predicted differences in trade probabilities explain one-way trade patterns.
- Predicted differences in extensive margins explain a meaningful share of bilateral trade imbalances among country pairs that trade in both directions.

Economic message: These tables are central for interpretation. The paper is not mainly a sample-selection paper; it is an extensive-margin paper. It also shows that directional asymmetries are not noise - they are systematically related to the model's predicted exporter-selection margins.

TABLE IV
BIAS DECOMPOSITION

Variables	1986 full sample			
	Benchmark	NLS	Firm heterogeneity	Heckman selection
Distance	-1.176** (0.031)	-0.798** (0.039)	-0.769** (0.038)	-1.214** (0.031)
Land border	0.458** (0.147)	0.834** (0.132)	0.855** (0.142)	0.436** (0.149)
Island	-0.391** (0.121)	-0.169 (0.120)	-0.164 (0.118)	-0.425** (0.120)
Landlock	-0.561** (0.188)	-0.447** (0.172)	-0.433* (0.187)	-0.565** (0.187)
Legal	0.486** (0.050)	0.387** (0.048)	0.381** (0.049)	0.488** (0.050)
Language	0.176** (0.061)	0.023 (0.062)	0.023 (0.060)	0.223** (0.061)
Colonial ties	1.299** (0.120)	1.001** (0.204)	0.979** (0.119)	1.311** (0.123)
Currency union	1.364** (0.255)	1.023** (0.273)	0.996** (0.260)	1.391** (0.257)
FTA	0.759** (0.222)	0.380* (0.182)	0.314+ (0.168)	0.737** (0.235)
Religion	0.102 (0.096)			
δ (from $\hat{w}_{ij}^*$)		0.871** (0.028)		
$\hat{\eta}_{ij}^*$		0.372** (0.069)		0.265** (0.070)
$\hat{z}_{ij}^*$			0.892** (0.051)	
Observations	11,146	11,146	11,146	11,146
R^2	0.709		0.716	0.710

Notes. m_{ij} is dependent variable throughout. Exporter and importer fixed effects. Religion is excluded variable in all second stage specifications. Bootstrapped standard errors for NLS; robust standard error (clustering by country pair) elsewhere.

*Significant at 10%.

**Significant at 5%.

***Significant at 1%.

(a) Table IV: Bias decomposition

TABLE V
ASYMMETRIES

Variable	1986 full sample			
	$T_{ij} - T_{ji}$	$m_{ij} - m_{ji}$		
		NLS	Polynomial	
$\hat{\rho}_{ij} - \hat{\rho}_{ji}$	0.999** (0.0169)			
$\hat{w}_{ij}^* - \hat{w}_{ji}^*$		1.187** (0.042)	1.251** (0.266)	
$\hat{v}(\hat{z}_{ij}^*) - \hat{v}(\hat{z}_{ji}^*)$				1.012** (0.035)
Country fixed effects	No	No	Yes	No
Observations	12,246	4,517	4,517	4,517
R^2	0.219	0.153	0.324	0.157

(b) Table V: Asymmetries

6.5 Table VI and Figure III: Heterogeneous trade-cost elasticities

Read:

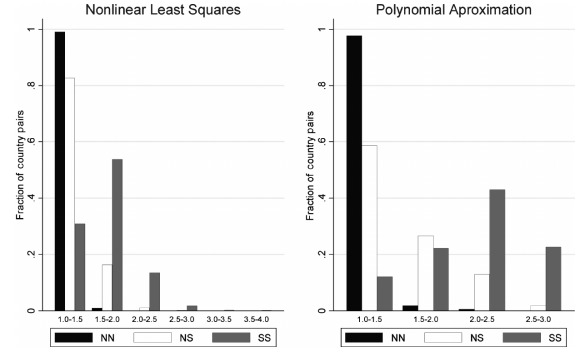
- Table VI reports summary statistics for the elasticity response to a distance-related trade-cost reduction.
- The intensive-margin elasticity is fixed within each specification, so variation in total elasticities comes from the extensive margin.
- North-North pairs have the lowest average elasticities, North-South pairs are higher, and South-South pairs are highest.
- In the NLS estimates, the average elasticity is about 1.29 for North-North pairs, 1.40 for North-South pairs, and 1.70 for South-South pairs.
- Figure III visualizes the distribution of distance elasticities across country-pair groups.
- The distribution shifts right as country pairs involve lower-income countries, especially South-South trade.

Economic message: Trade-cost reductions do not have a single universal effect. The extensive margin generates much larger responses for some country pairs than others, especially less-developed country pairs.

TABLE VI
SUMMARY STATISTICS OF THE TRADE ELASTICITY RESPONSE ACROSS COUNTRY PAIRS

Country pairs group	Number of country pairs	Nonlinear least squares				Polynomial approximation			
		Mean	S. D.	Min	Max	Mean	S. D.	Min	Max
NN	342	1.292	0.034	1.283	1.642	1.290	0.107	1.141	2.222
NS	4,626	1.404	0.152	1.283	2.949	1.526	0.386	1.141	2.895
SS	6,178	1.698	0.303	1.283	3.777	2.130	0.443	1.141	2.995
Overall	11,146	1.563	0.289	1.283	3.777	1.854	0.517	1.141	2.995

(a) Table VI: Elasticity responses across country pairs



(b) Figure III: Distribution of distance elasticities

7 Short summary

- The paper develops a gravity estimator for settings with heterogeneous firms, fixed export costs, and many zero bilateral trade flows.
- The model predicts that only sufficiently productive firms export to a destination.
- Some country pairs have no positive trade because even the most productive firm cannot cover the fixed cost of exporting.
- Aggregate trade flows depend on both trade per exporter and the number of exporters.
- The paper estimates a first-stage Probit for whether country j exports to country i .
- The second-stage trade-flow equation uses first-stage predictions to correct for sample selection and the endogenous extensive margin.
- Standard gravity estimates are biased because they omit the extensive-margin term.
- The economic importance of the bias comes mainly from the omitted extensive margin, not from the Heckman selection correction.
- Corrected estimates show that distance, FTAs, language, and other variables often have smaller direct effects on trade per exporter than benchmark gravity estimates imply.
- Some variables, such as language and religion, appear to affect trade mostly by lowering fixed costs and encouraging firms to enter export markets.
- The estimated extensive margin helps explain one-way trade and bilateral trade imbalances.
- Trade-cost elasticities vary substantially across country pairs, with larger responses for South-South pairs than for North-North pairs.

8 Conclusion

8.1 Core conclusion

Helpman, Melitz, and Rubinstein (2008) show that estimating trade flows requires jointly modeling who trades and how much they trade. In a world of heterogeneous firms and fixed export costs, zeros

are informative and positive trade volumes depend on the number of exporting firms. Standard gravity regressions that ignore these features overstate the direct effect of trade frictions.

8.2 Economic intuition

A trade barrier affects two margins. First, among firms that already export, it reduces trade per firm. Second, it raises the productivity cutoff required to export, so fewer firms enter the destination market. A standard gravity equation confounds these two effects. The paper’s estimator separates them by using the probability of positive trade to infer the extensive margin.

8.3 Incremental contribution revisited

The paper contributes a tractable empirical implementation of a Melitz-style heterogeneous-firm model using aggregate bilateral data. It extends gravity estimation to zero trade flows, one-way trade, and asymmetric trade volumes. It also shows how to estimate the extensive-margin contribution without directly observing firm-level exporters in every country.

8.4 Implications for empirical trade research

- Researchers should not automatically drop zero trade flows when estimating gravity equations.
- Variables that seem to increase trade volumes may operate mainly by increasing the number of exporting firms.
- Separate exporter and importer fixed effects are important when using directional bilateral trade data.
- Policy counterfactuals should account for extensive-margin responses, especially for developing-country trade.
- Sector-level trade data, where zeros are even more common, may benefit especially from this framework.

8.5 Limitations

- The empirical extensive margin is inferred from aggregate data rather than directly observed for all countries.
- The validity of the two-stage approach depends on credible excluded variables for fixed trade costs.
- The parametric version relies on distributional assumptions, though the paper shows robustness to less parametric approaches.
- The cross-sectional estimates do not directly identify all deep structural parameters, such as the elasticity of substitution, without additional restrictions.
- The approach is designed for long-run trade relationships and may not fully capture short-run dynamics of firm entry and exit.

8.6 Takeaway

The paper's main message is simple but powerful: trade frictions shape both the amount exported by existing exporters and the number of firms that export. Because the extensive margin is large, standard gravity estimates that ignore it can be substantially misleading.